

XJCO2913 CWK1 Report - Group 10

Part I (10 points)

Q1. The number of team members

The team consists of four members: Siyuan Jiang, Jiahui Zeng, Yuhao Lin, and Haozhe Peng.

Q2. The Belbin roles of every team member

According to Belbin's Team Roles theory, effective teams are formed through a balance of complementary behavioural contributions rather than technical ability alone (Belbin, 2010). In our team, **Siyuan Jiang** acts as the **Chair (CH)**. The Chair's function is to clarify objectives, coordinate contributions, and allocate work appropriately, which corresponds to Belbin's description of the role as a calm and objective coordinator who ensures that team resources are used effectively (Belbin, 2010). In our project context, Siyuan leads planning discussions, organises meetings, and ensures that responsibilities are clearly distributed.

Jiahui Zeng is the **Plant (PL)**. Belbin identifies the Plant as the role most associated with originality and idea generation, especially when a team needs novel solutions to complex problems (Belbin, 2010). Jiahui contributes primarily through proposing innovative features for scooter booking and location tracking, and through shaping user interface concepts that improve usability.

Yuhao Lin fulfils the **Monitor/Evaluator (ME)** role. This role is defined by critical judgement, logical reasoning, and impartial assessment of alternatives (Belbin, 2010). Yuhao evaluates our architecture, database choices, and testing strategy to ensure that key decisions are realistic and technically sound.

Finally, **Haozhe Peng** is the **Company Worker (CW)**, also commonly referred to as the Implementer in Belbin's framework. Belbin describes this role as disciplined and practical, with a focus on converting ideas into concrete, workable outputs (Belbin, 2010). Haozhe translates team decisions into implementation by developing core functionality and database models with attention to detail.

Q3. Team strengths (3 points)

Our core strength is the way our existing Belbin role combination supports the technical requirements of an electric scooter rental system. The **Plant** contributes creative feature ideas, while the **Company Worker** transforms those ideas into reliable implementation, particularly in backend and database development. The **Chair** maintains coordination and alignment across tasks, and the **Monitor/Evaluator** provides rigorous scrutiny of design and testing decisions. This complementary arrangement supports both innovation and quality control, enabling the team to produce solutions that are not only original but also feasible and robust (Belbin, 2010). Therefore, our principal strength is not a single individual skill, but a role-based collaboration structure that links idea generation, coordination, critical evaluation, and execution.

Q4. Team weaknesses (3 points)

A key behavioural weakness in our team is the absence of both a **Shaper (SH)** and a **Team Worker (TW)**. Belbin (2010) characterises the Shaper as the role that drives urgency, sustains momentum under pressure, and challenges complacency when progress slows. Without this role, the team may struggle to maintain pace during demanding periods, especially near sprint deadlines. At the same time, Belbin describes the Team

Worker as the role that supports interpersonal cohesion, mediates tensions, and helps maintain collaborative morale (Belbin, 2010). Without a Team Worker, communication gaps or emotional friction may remain unresolved for longer than is healthy for team performance.

This gap is particularly significant in our role composition because the **Monitor/Evaluator's** necessary critical feedback may unintentionally discourage the **Plant** or frustrate the **Company Worker** when not moderated through a diplomatic team-maintenance role. Consequently, the **Chair** may carry a disproportionate dual burden: both maintaining delivery momentum (a Shaper-like function) and preserving morale/relationship stability (a Team Worker-like function). Over time, this concentration of pressure on the Chair may reduce team resilience and increase coordination risk.

Q5. Mitigation strategy (4 points)

To mitigate the missing **Shaper**, the **Chair** will institutionalise short-interval accountability within our existing bi-weekly status meetings. Instead of broad progress updates only, each meeting will define narrowly scoped, time-bound checkpoints for every member, and completion of these checkpoints will be reviewed at the next session. This mechanism is designed to create constructive performance pressure and maintain continuity of progress between meetings, thereby reproducing the forward-driving effect that a Shaper would usually provide.

To mitigate the missing **Team Worker**, we will adopt a structured, blameless conflict-resolution protocol during Sprint Retrospectives. When disagreement occurs, the team will first frame the issue in neutral, evidence-based terms, then allow each member to explain perspective and constraints, and finally agree on specific improvement actions without personal attribution of fault. This protocol is intended to handle interpersonal friction objectively and preserve psychological safety, particularly when the **Monitor/Evaluator's** critiques are directed at ideas proposed by the **Plant** or implementation choices made by the **Company Worker**. In this way, team cohesion is protected through process design, even in the absence of a naturally relational Team Worker role.

References (Part I)

Belbin, R. M. (2010) *Team Roles at Work*. 2nd edn. Oxford: Butterworth-Heinemann.

Part II: GitHub Project Environment (5 Marks)

1. Repository Structure

XJCO2913_SE_Project

Public

Pin

Watch 0

Fork 0

Star 0

main

2 Branches

0 Tags

Go to file

Add file

Code

simple405

Merge pull request #4 from Darklord1243/chore/stack-prettier-guardrails

0b82246 · 1 hour ago

13 Commits

database	feat: modularize auth routes and migrate frontend to React w...	1 hour ago
docs	chore: initialize project structure and base files	8 hours ago
scripts	chore: initialize project structure and base files	8 hours ago
shared	chore: initialize project structure and base files	8 hours ago
src	feat: modularize auth routes and migrate frontend to React w...	1 hour ago
tests	chore: initialize project structure and base files	8 hours ago
.gitignore	chore: initialize project structure and base files	8 hours ago
.prettiignore	chore: establish project tech stack, prettier config, and contri...	5 hours ago
.prettierrc	chore: establish project tech stack, prettier config, and contri...	5 hours ago
CONTRIBUTING.md	chore: establish project tech stack, prettier config, and contri...	5 hours ago
README.md	feat: modularize auth routes and migrate frontend to React w...	1 hour ago
package-lock.json	feat: modularize auth routes and migrate frontend to React w...	1 hour ago
package.json	feat: modularize auth routes and migrate frontend to React w...	1 hour ago
vite.config.js	feat: modularize auth routes and migrate frontend to React w...	1 hour ago

About

No description, website, or topics provided.

Readme

Contributing

Activity

0 stars

0 watching

0 forks

Releases

No releases published

Create a new release

Packages

No packages published

Publish your first package

Contributors 3

Darklord1243

simple405

lingod

Caption: The repository structure is successfully set up for the whole team, separating the React frontend and Node backend.

2. Issue Tracker

Darklord1243 / XJCO2913_SE_Project

Q Type to search

Code Issues Pull requests Actions Projects Wiki Security and quality Insights Settings

Labels

New label

Search all labels

17 labels

backend

bug

database

documentation

duplicate

enhancement

feature

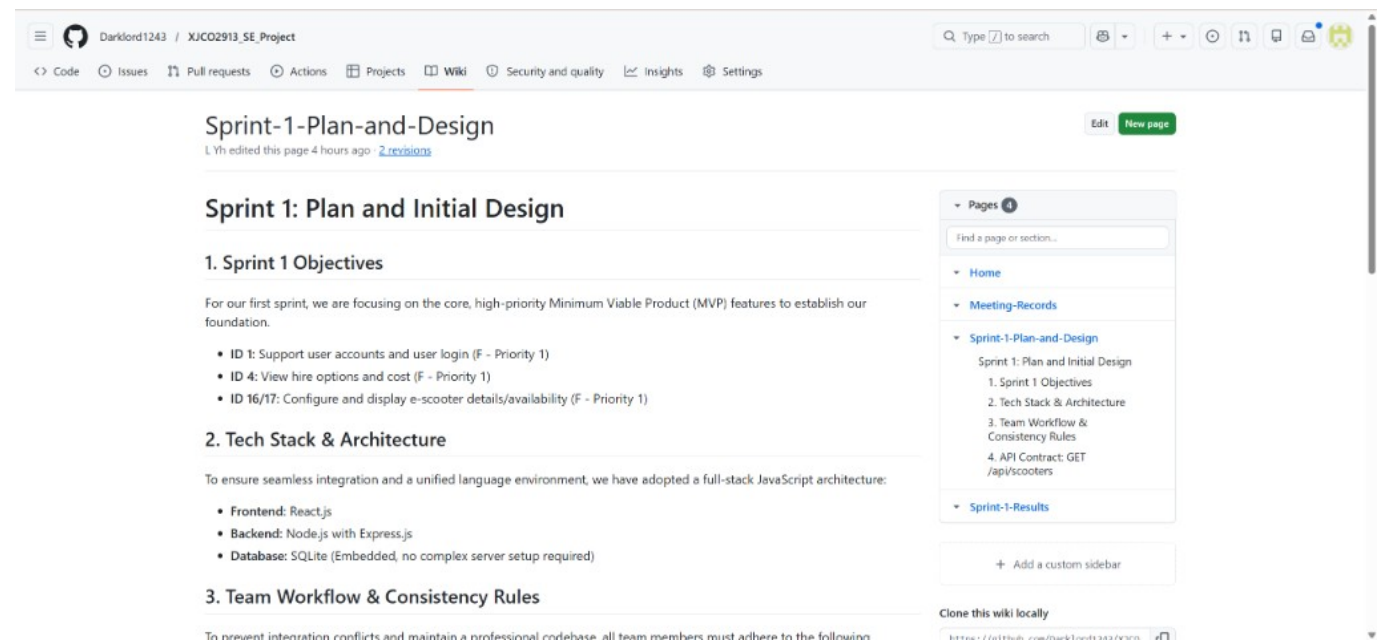
frontend

good first issue

Caption: The issue tracker is actively used, showing assigned tasks, priority labels, and closed issues for Sprint 1.

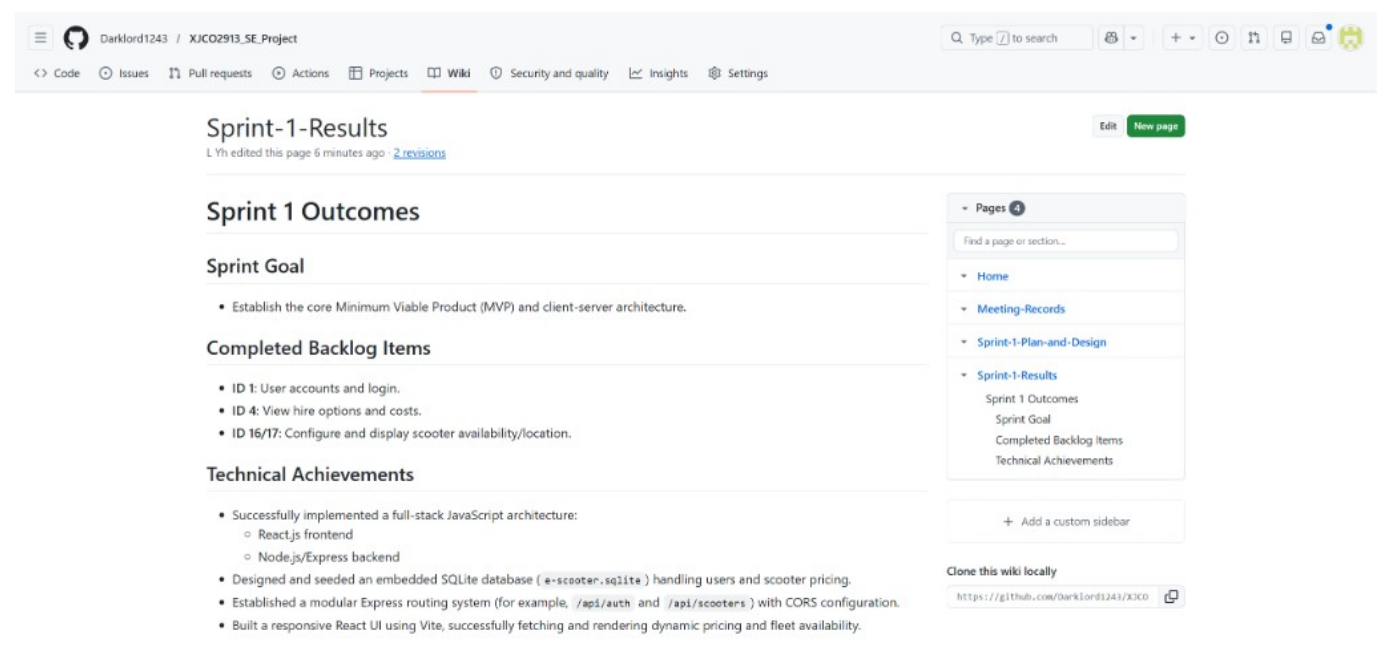
3. Sprint 1 Plan and Initial Design

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Caption: The Wiki documents our Initial Design, including our full-stack JavaScript architecture and SQLite database choice.

4. Sprint 1 Outcomes



Caption: The Wiki is actively updated with Sprint 1 outcomes, detailing the completion of IDs 1, 4, 16, and 17.